Reading coefficients and instrument diagnostics

Results: coefficient table and plot; Observations; County calibration observations; Instrument Wald F; County calibration instrument F; Stock–Wright LM S and p-value.

1. Coefficients and confidence intervals. The density coefficient is an elasticity: 0.05 corresponds to about 0.05% higher GDP per employment for a 1% rise in density, holding controls fixed. CH reports $\rho = \theta - 1$, so zero means $\theta = 1$. Standard errors appear in parentheses. The plot shows conventional 95% Wald confidence intervals, using the selected covariance method. Stars denote $p < 0.10$, $p < 0.05$ and $p < 0.01$. An interval containing zero means the data do not reject a zero coefficient at that level; it does not establish that the effect is zero. Conventional IV intervals are not robust to weak instruments.

2. Instrument Wald F. In ordinary IV, the first-stage regression predicts modern density or employment using the historical instrument and controls. With one excluded instrument, the reported statistic is

$$F = \left(\frac{\widehat{\pi}_z}{\mathrm{se}(\widehat{\pi}_z)} \right)^2,$$

where $\widehat{\pi}_z$ is its first-stage coefficient. The selected standard errors and final model sample are used. A small value indicates little evidence of conditional predictive power. The statistic does not test whether the instrument affects productivity through other channels.

For CH GMM, there is no ordinary first stage. Instead, the app uses how much the log index changes when θ changes: $\partial \log D_g(\theta) / \partial \theta$. It regresses this quantity on the historical instrument and controls. The resulting Wald F measures relevance near the fitted θ, treating that quantity as fixed. Conventional linear-IV cutoffs do not establish strong identification here. With county geography adjustments, this diagnostic also conditions on estimated geography. County calibration instrument F describes relevance in that county regression; it does not apply to Schooling and college.

3. Stock–Wright LM S. This CH IV statistic tests zero density elasticity, $H_0 : \rho = 0$. A small p-value is evidence against that hypothesis, assuming the instrument is valid for the model. The test uses a chi-squared reference distribution with one degree of freedom and remains valid with weak regional instruments under those assumptions. It is not a test of instrument strength. A large p-value means zero remains compatible with the data. For adjusted CH, the zero-effect model retains the selected adjustments. With geography, it includes geography-estimation uncertainty and requires reliable identification in the county regression.

4. Check the sample before comparing. Observations counts the regions actually estimated; County calibration observations counts the linked county sample when geography is included. Missing inputs, historical coverage and fixed effects can change these counts. Compare estimates on the same sample when assessing whether changing the instrument or adjustment changes the relationship. None of the reported diagnostics by itself establishes instrument validity.